

# THE STRUCTURE OF THE DEDEKIND ENDGAME: ONE APPROXIMATION STATEMENT AND TWO AUXILIARIES

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**ABSTRACT.** The comparison of the type and test model structures on the Dedekind cubical site was reduced, in the companion manuscript, to a single cubical approximation statement together with two combinatorial statements about the subatom order. We record a structural clarification that unifies the three: both combinatorial statements feed the *same* chain-approximation master statement. The first, well-foundedness of the subatom order, reduces to a tight bounded-generation conjecture (companion note). The second, intrinsic-chain-anchored asphericity, is shown here to be *equivalent* to chain approximation on the anchored subobjects of atoms: the intrinsic chain stratum of an anchored subobject  $W$  with  $\text{Ch}(A) \subseteq W \subseteq A$  satisfies  $\text{Ch}(W) = \text{Ch}(A)$ , so  $W$  is type-contractible iff its own chain approximation holds. Thus the Dedekind endgame has a single homotopical core—chain approximation—and two auxiliaries: a finiteness (securing the well-founded induction) and the anchored instance of the core itself. The anchored instance is test-verified in every computed case.

## 1. THE ENDGAME AFTER REDUCTION

In the companion manuscript the Dedekind type/test comparison is proved equivalent, through Cisinski regularity, a cosimplicial resolution and a singular Quillen adjunction, to a single *chain-approximation* statement: for every fibrant  $X$  the inclusion of the intrinsic chain stratum  $\text{Ch}(X) \hookrightarrow X$  is a type equivalence. The singular complex computes the *test* homotopy type of every presheaf, so  $\text{Ch}(X) \hookrightarrow X$  is always a test equivalence; only the type-level upgrade remains. An adversarially audited assembly then localizes that upgrade to two combinatorial statements:

- (a) the subatom order of  $\square^n$  is well-founded;
- (b) every subobject  $W$  of a cyclic atom  $A$  (or its fold and stabilizer quotients) with  $\text{Ch}(A) \subseteq W \subseteq A$  is type-contractible (*intrinsic-chain-anchored asphericity*).

We record how (a) and (b) both reduce to the chain-approximation core.

## 2. (A) REDUCES TO BOUNDED GENERATION

**Theorem 2.1** (companion note). *The measure  $|\text{im}|$  is non-increasing along the subatom order, and if for each finite lattice  $L$  there are finitely many atoms with image  $L$  then the order is well-founded. This finiteness follows from the tight conjecture that an atom is generated at arity at most the height of its image; the conjecture is verified, and tight, in every computed case.*

So (a) secures the well-founded induction that drives the assembly; it is a *finiteness*, orthogonal to the homotopy content.

## 3. (B) IS THE ANCHORED INSTANCE OF CHAIN APPROXIMATION

Write  $\text{Ch}(A)$  for the intrinsic chain stratum of an atom  $A$ : the sorted cells of  $A$  and their chain instances (equivalently the image of  $\text{Ch}_n$  under the sort action), which is a subpresheaf. A subobject  $W$  is *anchored* if  $\text{Ch}(A) \subseteq W \subseteq A$ .

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**Lemma 3.1.** *For an anchored subobject  $W$ ,  $\text{Ch}(W) = \text{Ch}(A)$ .*

*Proof.* Sortedness is a property of a cell itself, independent of the ambient object; hence the sorted cells of  $W$  are exactly the sorted cells of  $A$  that lie in  $W$ . Since  $\text{Ch}(A) \subseteq W$ , every sorted cell of  $A$  and every chain instance of one already lies in  $W$ , so the sorted cells of  $W$  are all of those of  $A$  and their chain instances again lie in  $W$ . Therefore  $\text{Ch}(W) = \text{Ch}(A)$ . (Machine-checked on the median atom and others: `dedekind_o32_anchor.py`.)  $\square$

**Proposition 3.2.** *Let  $A$  be a cyclic atom, so  $\text{Ch}(A) \hookrightarrow A$  is a type equivalence and  $\text{Ch}(A)$  is type-contractible. For an anchored  $W$ ,*

$$W \text{ is type-contractible} \iff \text{Ch}(W) \hookrightarrow W \text{ is a type equivalence.}$$

*Hence (b) holds iff chain approximation holds on every anchored subobject of every cyclic atom—a sub-case of the master statement.*

*Proof.* By Lemma 3.1,  $\text{Ch}(W) = \text{Ch}(A)$ , which is type-contractible. If  $\text{Ch}(W) \hookrightarrow W$  is a type equivalence then  $W \simeq \text{Ch}(W)$  is contractible. Conversely, if  $W$  is contractible then both  $W$  and  $\text{Ch}(W) = \text{Ch}(A)$  are contractible, so the inclusion between them is a type equivalence.  $\square$

**Corollary 3.3.** *For an anchored  $W$ , the inclusion  $\text{Ch}(W) \hookrightarrow W$  is always a test equivalence (the singular complex computes the test type, and  $\text{Ch}(W) = \text{Ch}(A)$  is test-contractible), so  $W$  is test-contractible. The content of (b) is exactly the type-level upgrade—the same type-versus-test gap as the master statement, restricted to anchored subobjects.*

Thus (b) is not an independent obstruction: it is chain approximation itself, evaluated on the anchored subobjects of atoms.

#### 4. THE UNIFIED PICTURE

**Master: chain approximation on all fibrant  $X \iff$  Dedekind comparison**

(a) well-foundedness  $\iff$  bounded generation (finiteness)

(b)  $\iff$  chain approximation on anchored  $W \subseteq \text{atoms}$

The Dedekind endgame therefore has a single homotopical core, chain approximation, and two auxiliaries of different character: (a) is a finiteness that makes the assembling induction well-founded, and (b) is the master statement’s own restriction to the anchored subobjects of atoms. The cone mechanism that contracts constant-free atoms does not extend to this restriction—it fails already on the median atom—so a non-cone, triangulation-level contraction is what remains.

#### 5. VERIFICATION

For the median atom (where the cone fails) and the constant-free five-chain, and across a sweep of sixteen atoms with  $|L| \in \{3, 4, 5\}$ , median- and non-median-fibre alike, every sampled anchored subobject  $W$  is test-contractible (mod-2 Betti 1, 0, 0);  $\text{Ch}(A)$  is a proper subobject in the majority of cases, so the tests are not vacuous. The scripts are `dedekind_o32_conechain.py` (cone (non-)preservation of the intrinsic stratum) and `dedekind_o32_anchor.py` (anchored- $W$  Betti sweep), in the artifact of the companion manuscript.

#### REFERENCES

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